# SFWRENG 3DX4 - Pre-lab 2

Matthew Farah, ⟨farahm11@mcmaster.ca⟩, 400468251

February 5, 2026

## Contents

**Question 1** **2**
Part 1.a . . . . . 2
Part 1.b . . . . . 4
Part 1.c . . . . . 6
Part 1.d . . . . . 7

**Question 2** **8**

**Question 3** **9**

**Question 4** **10**

**Question 5** **11**

**Question 6** **13**

## Question 1

Once again, our first order approximation of the transfer function of a DC electric motor with respect to angular velocity is

$$G_\omega = \frac{\Omega(s)}{V(s)} = \frac{A}{\tau s + 1}$$

Where $A$ and $\tau$ are positive, real-valued constants. $\Omega(s)$ and $V(s)$ are angular velocity and voltage as functions of $s$ in the Laplace domain. (Note: $\Omega$ is capital $\omega$ so we are using $\Omega(s) := L\{\,\omega(t)\,\}$.)

We will now develop a formula for the motor DC gain constant $A$ in terms of a step input change $\Delta V$ and step output change $\Delta\omega$.

**(a)** Using the final value theorem, find an expression for the steady state value of $\omega(t)$ when a step input of amplitude $V_x$ is applied. NOTE: $\tau > 0$ so the pole of $G_\omega(s)$ at $s = \frac{-1}{\tau}$ is in the open Left Half Plane (LHP) so the system is stable! Therefore, we can use the final value theorem.

**Solution:**

We know that the steady state value of $\omega(t)$ is given by $\lim_{t\to\infty}\omega(t)$. By the final value theorem, we know that $\lim_{t\to\infty}\omega(t) = \omega(\infty) = \lim_{s\to 0} s\Omega(s)$. Since $\frac{\Omega(s)}{V(s)} = \frac{A}{\tau s+1}$, we therefore know that:

$$\Omega(s) = \frac{AV(s)}{\tau s + 1}$$

And so:

$$s\Omega(s) = \frac{sAV(s)}{\tau s + 1}$$

Now, note that a step input of amplitude $V_x$ is given by $V_x u(t)$. By the linearity of the Laplace transform and the lookup table, we have that $\mathcal{L}\{\,V_x u(t)\,\} = \frac{V_x}{s}$, so:

$$\begin{aligned} s\Omega(s) &= \frac{sAV_x}{s(\tau s + 1)} \\ &= \frac{AV_x}{\tau s + 1} \end{aligned}$$

Hence:

$$\begin{aligned}
\lim_{s\to 0} s\Omega(s) &= \lim_{s\to 0} \frac{AV_x}{\tau s + 1} \\
&= \frac{AV_x}{\tau(0) + 1} \\
&= \frac{AV_x}{1} \\
&= AV_x
\end{aligned}$$

**(b)** Give the expression for $\omega(t)$ in response to a step input $V_x$ at time $t = 0$. Assume a non-zero initial condition for $\omega(t)$, i.e., $\omega(0) = \omega_0$. Note that the response due to a non-zero initial condition $\omega_0$ can be modeled as the response due to input $v(t) = \omega_0\delta(t)$ where $\delta(t)$ in the impulse response. Since $G_\omega(s)$ is a linear system, the response to step input $V_x$ with non-zero initial condition $\omega_0$ is just the sum of the responses due to the step and the initial condition.

**Solution:**

To find an expression for $\omega(t)$, we must find $\mathcal{L}^{-1}\{\,\Omega(s)\,\}$. Since we are given that $\omega(0) = \omega_0$ is non-zero, we can use the linearity of the transfer function $G_\omega(s)$ to find $\Omega(s)$. Since the input to the system $v(t)$ including both the initial condition and step input can be represented by:

$$v(t) = \delta(t)\omega_0 + V_x u(t)$$

We can use the linearity of the Laplace transform, our given conversion table, and the definition of the Dirac-delta function ($\delta(t)$) to conclude that:

$$V(s) = \omega_0 + \frac{V_x}{s}$$

Therefore, we have that:

$$\begin{aligned} G_\omega(s) = \quad \frac{\Omega(s)}{V(s)} \quad &= \frac{A}{\tau s + 1} \\ = \frac{\Omega(s)}{\omega_0 + V_x/s} &= \frac{A}{\tau s + 1} \end{aligned}$$

And so:

$$\begin{aligned} \Omega(s) &= \frac{A(\omega_0 + V_x/s)}{\tau s + 1} \\ &= \frac{A\omega_0}{\tau s + 1} + \frac{AV_x}{s(\tau s + 1)} \\ &= \frac{A\omega_0}{\tau}\left(\frac{1}{s + 1/\tau}\right) + AV_x\left(\frac{1}{s(\tau s + 1)}\right) \end{aligned}$$

Using linearity and the conversion table we are able to find the inverse Laplace of the term on the left. However, we must perform partial fraction decomposition on the rightmost term before we are able to take the inverse Laplace of $\Omega(s)$. Therefore, let $B, C$ be constants we will soon find such that:

$$\frac{1}{s(\tau s+1)} = \frac{B}{s} + \frac{C}{\tau s+1}$$

We therefore know that:

$$\begin{aligned} 1 &= B(\tau s+1) + Cs \\ &= B\tau s + B + Cs \end{aligned}$$

By equating coefficients, we can clearly see that $B = 1$, implying that $C = -\tau$. Substituting back into our original equation, we therefore have that:

$$\begin{aligned} \Omega(s) &= \frac{A\omega_0}{\tau}\left(\frac{1}{s+1/\tau}\right) + AV_x\left(\frac{1}{s} - \frac{\tau}{\tau s+1}\right) \\ &= \frac{A\omega_0}{\tau}\left(\frac{1}{s+1/\tau}\right) + AV_x\left(\frac{1}{s}\right) - AV_x\tau\left(\frac{1}{\tau s+1}\right) \\ &= \frac{A\omega_0}{\tau}\left(\frac{1}{s+1/\tau}\right) + AV_x\left(\frac{1}{s}\right) - \frac{AV_x\tau}{\tau}\left(\frac{1}{s+1/\tau}\right) \\ &= \frac{A\omega_0}{\tau}\left(\frac{1}{s+1/\tau}\right) + AV_x\left(\frac{1}{s}\right) - AV_x\left(\frac{1}{s+1/\tau}\right) \end{aligned}$$

And so, using the linearity of the inverse Laplace and the conversion table, we can conclude that:

$$\begin{aligned} \omega(t) &= \frac{A\omega_0}{\tau}e^{-\frac{1}{\tau}t} + AV_x u(t) - AV_x e^{-\frac{1}{\tau}t} \\ &= \frac{A(\omega_0 - V_x\tau)}{\tau}\left(e^{-\frac{1}{\tau}t}\right) + AV_x u(t) \end{aligned}$$

**(c)** For $\omega(t)$ you computed in 1.b.), what is $\lim_{t\to\infty}$? How does it compare to your answer from 1.a.)?

**Solution:**

Since $\tau > 0$, we know $-\frac{1}{\tau} < 0$ and hence:

$$\begin{aligned}
\lim_{t\to\infty} \omega(t) &= \lim_{t\to\infty} \frac{A(\omega_0 - V_x\tau)}{\tau}\left(e^{-\frac{1}{\tau}t}\right) + AV_x u(t) \\
&= \frac{A\omega_0}{\tau}(0) + AV_x(1) \\
&= AV_x
\end{aligned}$$

This is identical to the result obtained in 1.a.), implying that the initial state of $\omega(t)$ at $t = 0$ has no impact on the steady state value of $\omega(t)$ when the system's input is the same step function.

**(d)** Now assume that you run the motor with an initial step input of $V_{min}$ until time $t_0$ when the system has reached steady state before the step input is changed to $V_{max}$. In other words, your system input will take the form:

$$v(t) = \begin{cases} V_{max}, \ t \geq t_0 \\ V_{min}, \ 0 \leq t < t_0 \end{cases}$$

Where $t_0 >> \tau$ and $V_{min}$ and $V_{max}$ may be non-zero. Use the results above to show that:

$$A = \frac{\Delta\omega}{\Delta V}$$

Where:

- $\Delta V = V_{max} - V_{min}$
- $\Delta\omega = \omega_{ss} - \omega_0$, where $\omega_0$ is the steady state response to a constant input $V_{min}$ and $\omega_{ss}$ is the steady state response to the input $V_{max}$.

**Solution:**

By questions 1.a.), we know that:

$$\omega_0 = AV_{min}$$

And:

$$\omega_{ss} = AV_{max}$$

Which means:

$$\begin{aligned} \omega_{ss} - \omega_0 &= AV_{max} - AV_{min} \\ &= A(V_{max} - V_{min}) \end{aligned}$$

And hence:

$$A = \frac{\omega_{ss} - \omega_0}{V_{max} - V_{min}}$$

$$= \frac{\Delta\omega}{\Delta V}$$

## Question 2

Using the formula derived in question 1, use the following graphs to calculate $A$ for this system.

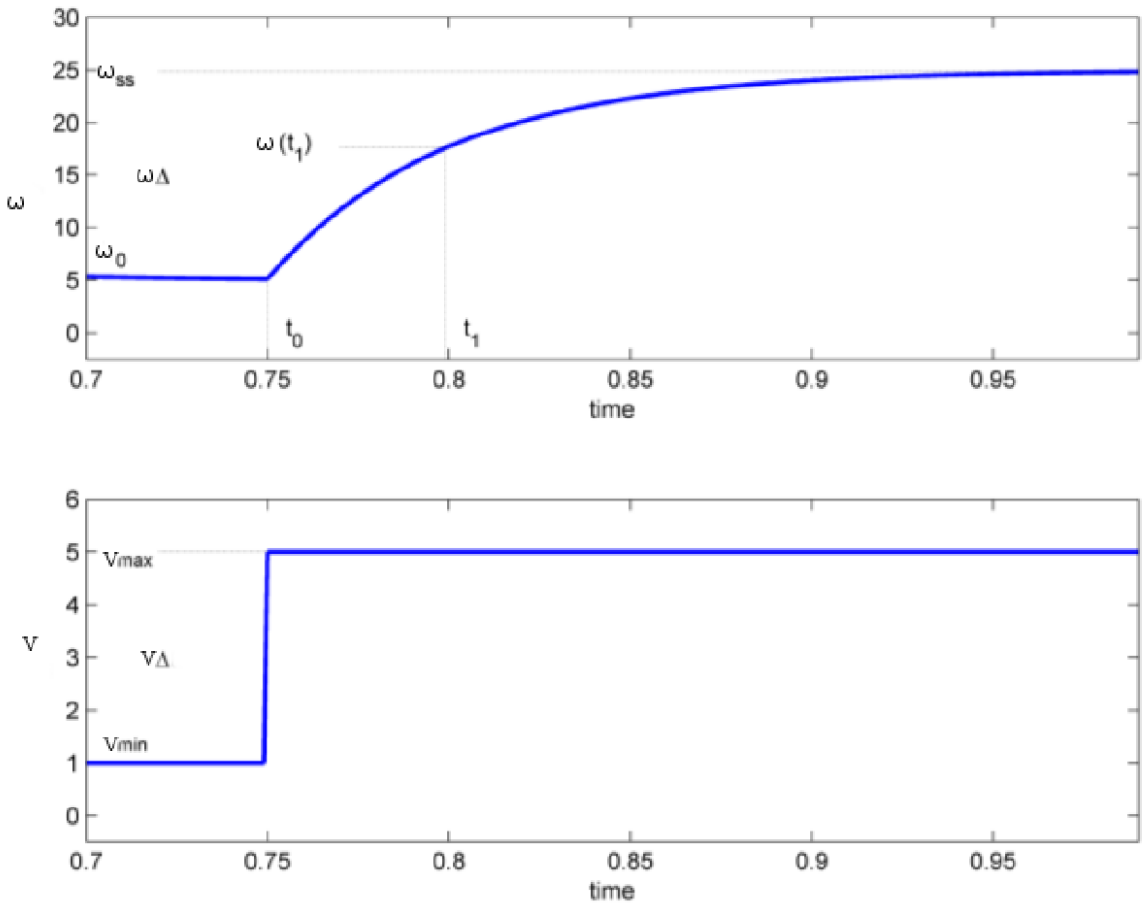


Figure 1: A First Order Response to Step Input

## Solution:

From the graph, we can clearly observe that $\Delta\omega = 20$ and $\Delta V = 4$. Thus, using the formula developed in 1.d.), we have:

$$A = \frac{\Delta\omega}{\Delta V} = \frac{20}{4} = 5$$

### Question 3

For a first order system, the time it takes a step response to reach 63.2% of its steady state value ($t_1 - t_0$ in Fig. 1) is the response's time constant $\tau$ . i.e., at time $t_1$, $\omega(t_1) = 0.632\Delta\omega + \omega_0$. Find the time constant $\tau$ for the above system in Fig. 1.

### Solution:

Reading off the graph, we can infer that $t_0 \approx .75$ and $t_1 \approx .80$. Hence:

$$\tau = t_1 - t_0 \approx .80 - .75 \approx .05$$

**Question 4**

Using $A$ and $\tau$, give the transfer function $G_\omega(s)$ in terms of $s$.

**Solution:**

Substituting for the values of $A$ and $\tau$ we found earlier, we have:

$$\begin{aligned} G_\omega &= \frac{A}{\tau s + 1} \\ &= \frac{5}{.05s + 1} \end{aligned}$$

### Question 5

Complete problem 21(a) in Chapter 4 of Control Systems Engineering 8th ed:
For the unit step response shown, find the transfer function of the system.

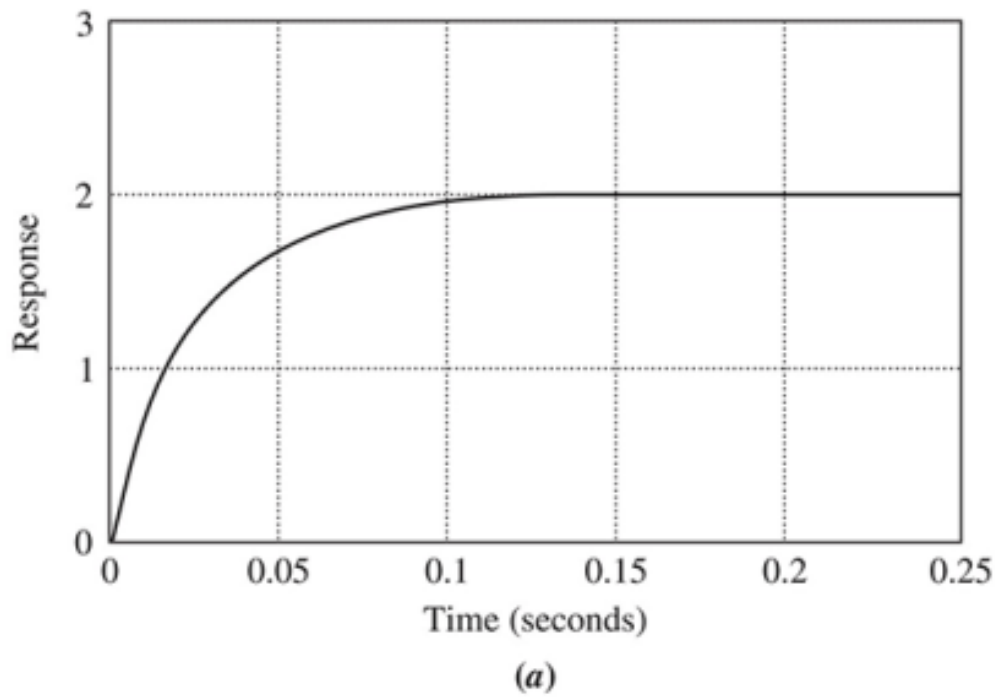


Figure 2: System response to step input

### Solution:

Since the graph provided represents the system's response to a unit step output, we know that $\Delta V = 1$. Reading from the graph, we can infer that the steady state of $\omega = 2$ and hence $\Delta\omega = 2$. Hence, we know that:

$$A = \frac{\Delta\omega}{\Delta V} = \frac{2}{1}$$

Once again, noting that the steady state of $\omega = 2$, we know that $.632\omega = 1.264$. We therefore know that $\omega(\tau) = 1.264$. Reading off the graph, we can therefore see that:

$$\omega(\tau) = 1.264 \implies$$
$$\tau \approx 0.025$$

Therefore:

$$
\begin{aligned}
G(s) &= \frac{A}{\tau s + 1} \\
&= \frac{2}{(0.025)s + 1}
\end{aligned}
$$

### Question 6

In one or two paragraphs, describe a situation in which it is preferable to derive a transfer function experimentally, and then design/calibrate a controller using simulation software, rather than experimenting with the plant to design/calibrate the controller directly. Describe a situation in which this is not preferable. (Answer in complete sentences!)

**Solution:**

Transfer functions might be easier to derive experimentally for a number of different reasons. Deriving the transfer function of a system experimentally involves testing the equipment over multiple iterations and taking averages of the observed time constants and calculated $A$ values and so, in the case the equipment isn't particularly delicate or rough approximations of the transfer function are acceptable, deriving it experimentally may likely save on time compared to attempting to find it analytically.

On the other hand, if the equipment being worked with is delicate, a more precise transfer function is needed, and there is enough time it may be preferable to first find the transfer function mathematically. Although mathematically finding and then experimentally verifying the transfer function of a system may take longer, doing so provides a reasonable baseline as t how the system will behave, greatly reducing the possibility the equipment is damaged by attempting to find the transfer function experimentally without a good baseline to work off. As well, a mathematical expression (accounting for real-world factors and assumptions) would likely also prove to be more accurate than what could be determined, unless a very large number of experiments are done. Therefore, if the system is particularly delicate, or a much more precise expression for the transfer function is required, it is likely preferable to attempt to derive it mathematically.